Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior backend engineer with 6+ years building and operating **event-driven microservices** in production. Architected a **Kafka**-based microservices backend on **Docker and Kubernetes**; built a redelivery-safe, **idempotent** two-way MongoDB↔Firestore sync on Google **Pub/Sub** that carried live traffic through a **zero-downtime** migration of 20+ applications on **GCP**. Owns the full software lifecycle — from design through deployment, production and decommissioning of legacy services (all workloads moved off AWS). Shipped production backends in Node.js, Python and Go. Based in Kampala (UTC+3), working with distributed teams across US, Singapore and Uganda time zones since 2020. Uses AI coding tools, including Claude Code, in day-to-day engineering. Primary stack is Node.js; keen to apply the same event-driven patterns in C#/.NET.

CORE COMPETENCIES

Event-Driven Microservices | Messaging: Kafka, Pub/Sub, NATS | Idempotency & Retry Design | Kubernetes & Docker | GCP & AWS Cloud | Full Lifecycle Ownership | Zero-Downtime Migrations | AI Coding Tools (Claude Code)

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer · US Remote

- Led the end-to-end migration of **20+ production applications** from Parse/MongoDB to Firebase/GCP with **zero downtime**, owning design, implementation, deployment and production iteration; reported directly to the CTO.
- Built a real-time, two-way MongoDB↔Firestore sync service in **Node.js** on Google **Pub/Sub**, designed for **idempotent processing, redelivery and out-of-order events** so both datastores could serve live traffic in parallel.
- Consolidated all services off AWS onto GCP (file storage S3→GCS via a Python pipeline), cutting infrastructure cost by **$5,000/month**.
- Shipped revenue features end-to-end: Express Shipping (**+$40 revenue per order**) and a 3D visualisation feature (video overlay + ffmpeg) that increased buyer conversion.

MTailor

Mar 2022 – Jul 2022

Backend Contractor · US Remote

- Ramped onto an unfamiliar Firebase/GCP stack and delivered the MongoDB→Firestore migration scripts with complex transformation logic; converted to full-time and given ownership of the whole migration.
- Wrote the Firebase SDK documentation (Node.js and Python) other engineers onboarded from; trained the Ops team on the new dashboard and backup procedures.

Mind2matter

Jan 2022 – Jul 2022

Software Engineer · US Remote

- Built backends for DeFi applications using **Node.js** and Web3; delivered React UIs from Figma designs under tight timelines, reporting to the CTO.

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer · Singapore Remote

- Built scheduled jobs querying **2M+ Firestore records** to keep customer-facing stock prices current from Morningstar APIs; extended backend APIs in **TypeScript** on Firebase Cloud Functions and Express.js.

CodeBits

Jan 2020 – Jul 2021

Software Engineer / Team Lead · Uganda Remote

- Architected an **event-driven microservices** backend for FIDA Uganda's case-management platform on **Apache Kafka, Docker and Kubernetes**, with NATS Streaming for service messaging.
- Built a **USSD service for legal-aid providers** talking to a Node.js backend over **gRPC** — reaching feature-phone users across Uganda.
- Led a team of 4 developers; delivered a paralegal database for LASPNET Uganda and a cross-platform React Native (Expo) app with Firebase push notifications.

PROJECTS

Event-Driven Microservices — FIDA Uganda

Kafka-based event-driven backend for a legal case-management platform, containerised and orchestrated on Kubernetes; services consumed by web, mobile and USSD clients.

Apache Kafka · Docker · Kubernetes · gRPC · NATS Streaming · Node.js

Real-time Pub/Sub Sync Pipeline

Two-way MongoDB↔Firestore sync with idempotent, redelivery-safe message handling, running under parallel production traffic across 20+ applications for the full migration window.

Node.js · Google Pub/Sub · MongoDB · Firestore

Healthcare WhatsApp Chatbot

Conversational assistant on the WhatsApp Business API integrating OpenAI and Google Vision for prescription OCR.

Node.js · OpenAI API · Google Vision · WhatsApp Business API

EDUCATION

B.Sc. Computer Engineering — Makerere University 2018 – 2022

Kampala, Uganda

LINKS

Open-source & project code — github.com/zac-09

SKILLS

Languages: JavaScript / Node.js, TypeScript, Python, Go (intermediate), Rust (intermediate)

Messaging & Streaming: Apache Kafka, Google Pub/Sub, NATS Streaming, gRPC

Cloud & Infra: GCP (Firestore, Cloud Functions, Pub/Sub, GCS), AWS (S3, EC2, EBS — intermediate), Docker, Kubernetes, Linux

Databases: MongoDB, Firestore **Tooling:** Git, Claude Code and AI-assisted development workflow **Spoken:** English, Kiswahili